

● HARD

● ADVANCED MATH

Advanced Math

18

7 questions · ~11 min

SAT QUESTION BANK

2026-05-29

Q1

ADVANCED MATH

$$fx = 4x^2 + 64x + 262$$

The function g is defined by $gx = fx + 5$. For what value of x does gx reach its minimum?

- A. -13
- B. -8
- C. -5
- D. -3

$$Pt = 2901.04\left(\frac{4}{5}\right)^t$$

The function P models the population, in thousands, of a certain city t years after 2005. According to the model, the population is predicted to increase by $n\%$ every 18 months. What is the value of n ?

- A. 0.38
- B. 1.04
- C. 4
- D. 6

An auditorium has seats for 1,800 people. Tickets to attend a show at the auditorium currently cost \$ 4.00. For each \$ 1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

- A. 4
- B. 7
- C. 14
- D. 18

Square P has a side length of x inches. Square Q has a perimeter that is 176 inches greater than the perimeter of square P. The function f gives the area of square Q, in square inches. Which of the following defines f ?

A. $f(x) = x + 44^2$

B. $f(x) = x + 176^2$

C. $f(x) = 176x + 44^2$

D. $f(x) = 176x + 176^2$

A machine launches a softball from ground level. The softball reaches a maximum height of 51.84 meters above the ground at 1.8 seconds and hits the ground at 3.6 seconds. Which equation represents the height above ground h , in meters, of the softball t seconds after it is launched?

- A. $h = -t^2 + 3.6$
- B. $h = -t^2 + 51.84$
- C. $h = -16t - 1.8^2 - 3.6$
- D. $h = -16t - 1.8^2 + 51.84$

x	p(x)
-2	5
-1	0
0	-3
1	-1
2	0

The table above gives selected values of a polynomial function p . Based on the values in the table, which of the following must be a factor of p ?

- A. $(x - 3)$
- B. $(x + 3)$
- C. $(x - 1)(x + 2)$
- D. $(x + 1)(x - 2)$

$$f(x) = x^2 - 44x - 46$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

- A. 46
- B. 45
- C. 44
- D. -1